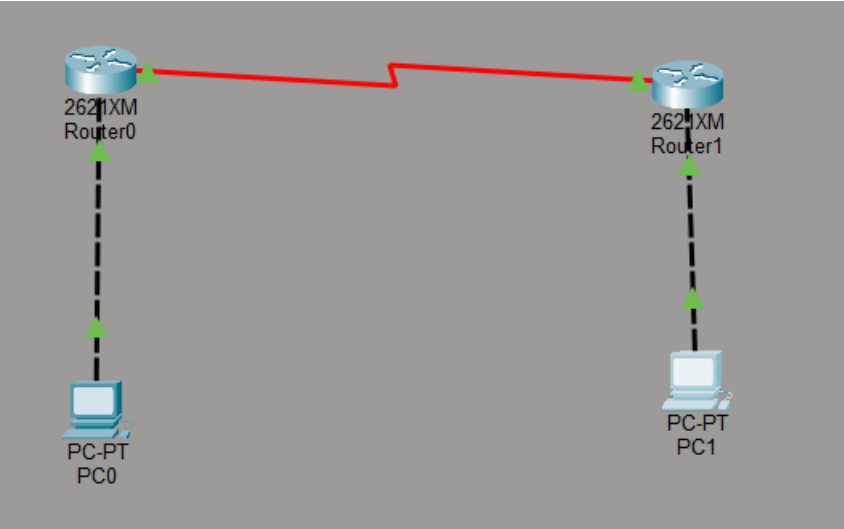


Q-On Router1 in Cisco Packet Tracer, configure a static route so it can reach a PC connected to Router2’s LAN network. Test connectivity using the ping command from Router1 to the PC.

The objective of this practical is to configure a static route on **Router1** so that it can communicate with a PC connected to the LAN of **Router0**. Connectivity is verified using the ping command.

TOPOLOGY



IP Addressing Table

Device	Interface	IP Address	Mask
Router0	Fa0/0	192.168.1.1	255.255.255.0
PC0	Fa0	192.168.1.10	255.255.255.0
Router0	Serial0/0	10.0.0.1	255.255.255.252
Router1	Serial0/0	10.0.0.2	255.255.255.252
Router1	Fa0/0	192.168.2.1	255.255.255.0
PC1	Fa0	192.168.2.10	255.255.255.0

Router0

```
Router0
Physical Config CLI Attributes

Press RETURN to get started!

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface serial 0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0, changed state to down
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#show ip interface brief
      ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          192.168.1.1     YES manual up                up
FastEthernet0/1          unassigned      YES unset  administratively down down
Serial0/0                 10.0.0.1        YES manual down              down
Router#
%LINK-5-CHANGED: Interface Serial0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

Router#ping 192.168.1.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

Router1

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 0/0
      ^
% Invalid input detected at '^' marker.

Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#interface serial 0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0, changed state to up

Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/7 ms
```

PC Configuration

PC0

IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

PC1

IP Address: 192.168.2.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

Static Route Configuration

```
Router#enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#do wr
^
% Invalid input detected at '^' marker.

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Serial0/0
S       192.168.1.0/24 [1/0] via 10.0.0.1
C       192.168.2.0/24 is directly connected, FastEthernet0/0

Router#
```

Connecting Testing

Router1# ping 10.0.0.1

The connection was successful.

Next, Router1 was used to test connectivity to PC0:

Router1# ping 192.168.1.10

successful output:

!!!!

Success rate is 100 percent (5/5)

This confirms that Router1 can successfully reach PC0 through Router0 using the configured static route.